Advanced Laser & Optics Learning Lab

Lab Record & Viva-Voce Booklet

15 Experiments — printable, fill-by-hand observation, calculation, result and viva sheets

Companion to the interactive simulator and online User Guide

Student Name: ____

Roll / Enrolment No.: ____

Course & Section: ____

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How to Use This Booklet

This booklet mirrors the 15 experiments in the online **User Guide** and the interactive simulator. For each experiment: run the simulator panel as instructed in the guide, record your readings directly in the observation table below, plot your graph on the provided 15×15 grid (choose your own scale per square), complete the calculation and result sections, work out the maximum permissible error using the least counts of the instruments you used, and prepare written answers to the viva-voce questions before your practical examination. Full theoretical background, governing formulas and step-by-step procedure for every experiment are available in the online guide — this booklet intentionally contains only the record-keeping sheets, not the theory, to keep it compact for printing.

Contents

1. Young's Double-Slit Interference

Aim

To study the interference pattern produced by two coherent slits and to determine the fringe width for a given wavelength, slit separation and screen distance.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$\beta = \lambda D / d$

β = fringe width measured on the screen (mm)
λ = wavelength of the laser light (nm)
D = distance from the double slit to the screen (m)
d = separation between the two slits (mm)

Labelled diagram of the experimental apparatus:

Observation Table

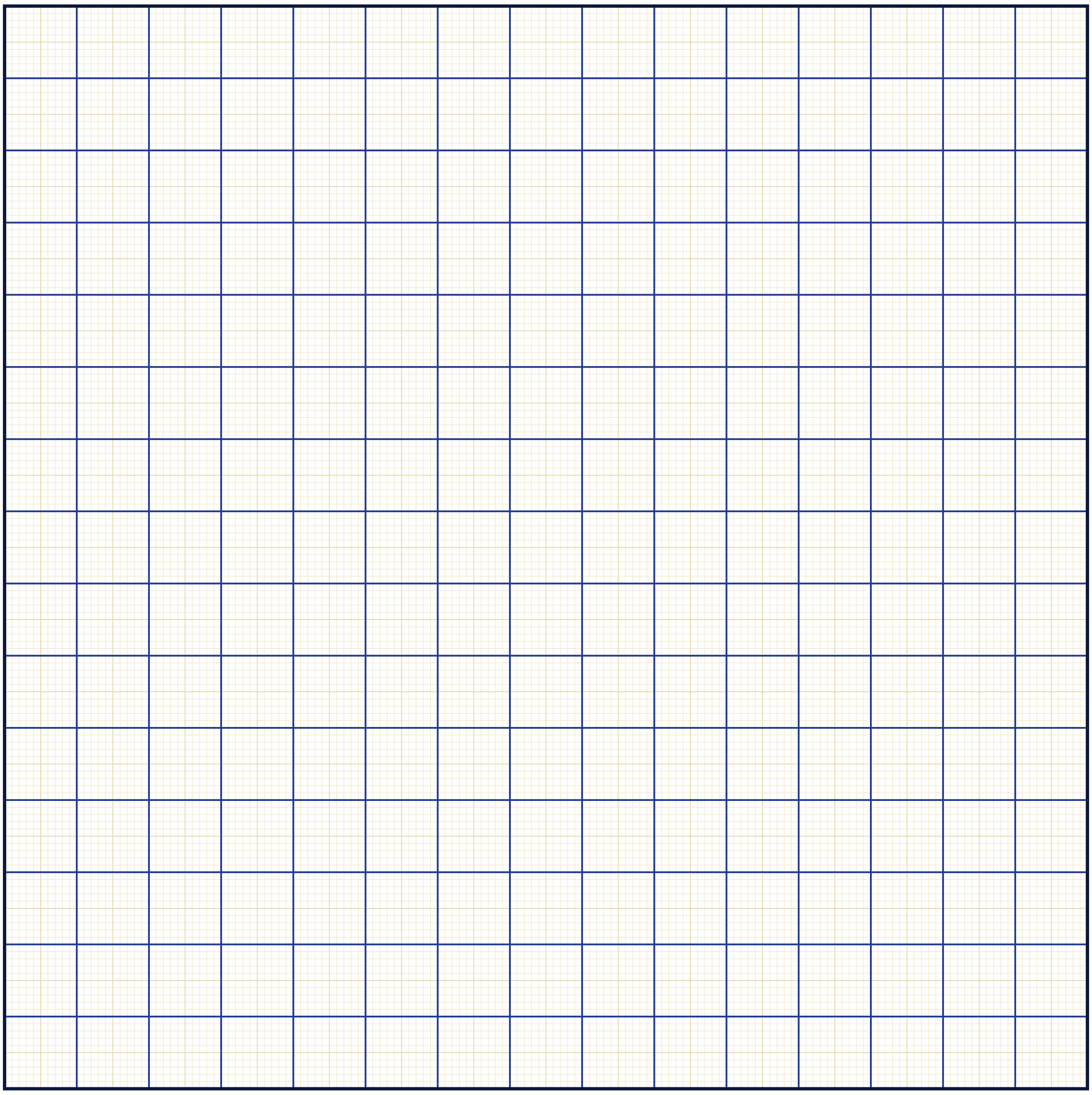
S. No.	λ (nm)	d (μm)	D (cm)	Fringe width β (mm) — observed	β = λD/d (mm) — calculated

S. No.	λ (nm)	d (µm)	D (cm)	Fringe width β (mm) — observed	β = λD/d (mm) — calculated

Graph

Fringe Width vs Wavelength

Fringe width, β (mm)



Wavelength, λ (nm)

Calculation

Result

Quantity varied	Slope from graph	Expected relation	% deviation

Maximum Permissible Error

Formula: $\Delta\beta / \beta = \Delta\lambda/\lambda + \Delta D/D + \Delta d/d$

$\beta = \lambda D/d$ is a simple product/quotient of three measured quantities, so the maximum fractional error is the sum of the fractional errors (least count / measured value) of each.

Quantity	Measured Value	Least Count (LC)
λ — Wavelength (nm)		
D — Slit-to-screen distance (cm)		
d — Slit separation (μm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why must the two sources be coherent for a stationary interference pattern to be observed?

2. What happens to the fringe width if the entire apparatus is immersed in a medium of refractive index n?

3. Distinguish between the interference pattern and the diffraction envelope that modulates it.

4. How would the fringe width change if the wavelength of the laser source were doubled, all else unchanged?

5. What happens to the fringe pattern if the separation between the two slits, d, is increased while D is kept fixed?

6. Why is a laser preferred over an ordinary sodium lamp for obtaining a clean, high-contrast fringe pattern in this setup?

7. What would you observe on the screen if one of the two slits were completely blocked?

2. Single-, Double- and N-Slit Diffraction

Aim

To observe how the diffraction pattern evolves from a single slit through a double slit to an N-slit grating, and to verify the grating equation and resolving power relation.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$a \sin\theta = m\lambda$$

a = width of the single slit (µm)
θ = angle of diffraction at which a minimum is observed
m = order of the diffraction minimum (m = ±1, ±2, ...)
λ = wavelength of the light used (nm)

Labelled diagram of the experimental apparatus:

Observation Table

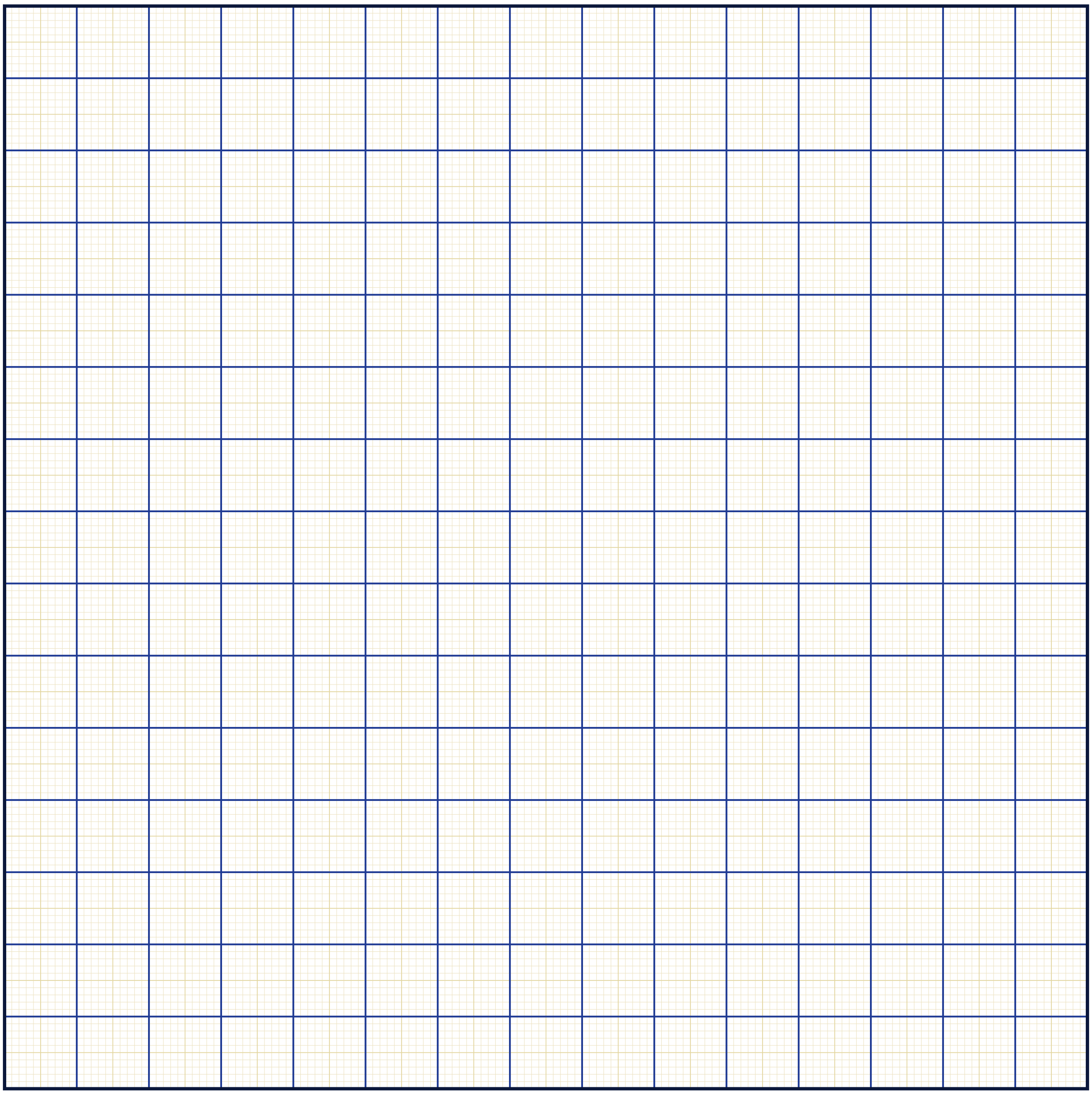
S. No.	N	a (µm)	(a+b) (µm)	λ (nm)	R = nN (n=1)

S. No.	N	a (µm)	(a+b) (µm)	λ (nm)	R = nN (n=1)

Graph

Grating Element vs sinθ (slope method)

Grating element, (a+b) (µm)



sinθ (order n = 1)

Calculation

Result

N	R (calculated)	R (from slope of graph)	% deviation

Maximum Permissible Error

Formula: $\Delta(a+b) / (a+b) = \Delta\lambda/\lambda + \cot\theta \cdot \Delta\theta$

From $(a+b) \sin\theta = n\lambda$, differentiating gives a $\cot\theta$ term for the angle uncertainty in addition to the wavelength's fractional error.

Quantity	Measured Value	Least Count (LC)
λ — Wavelength (nm)		
θ — Diffraction angle (°)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why do secondary maxima become negligible once N is large?

2. How does the resolving power of a grating relate to the total number of illuminated grooves rather than the groove spacing alone?

3. How does the width of the central diffraction maximum change if the slit width a is reduced?

4. Why are the secondary maxima in single-slit diffraction much weaker than the central maximum?

5. What is the physical difference between the diffraction envelope and the interference fringes seen in a double-slit pattern?

6. How does increasing the number of slits N change the sharpness of the principal maxima in grating diffraction?

7. Define the resolving power of a diffraction grating and state the factors it depends on.

3. Polarization & Malus's Law

Aim

To verify Malus's Law by measuring the transmitted intensity through an analyzer as a function of the angle between the polarizer and analyzer transmission axes.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$I(\theta) = I_0 \cos^2\theta$

$I(\theta)$ = intensity transmitted through the analyzer
I_0 = intensity after the first polarizer ($\theta = 0$)
θ = angle between the polarizer and analyzer transmission axes

Labelled diagram of the experimental apparatus:

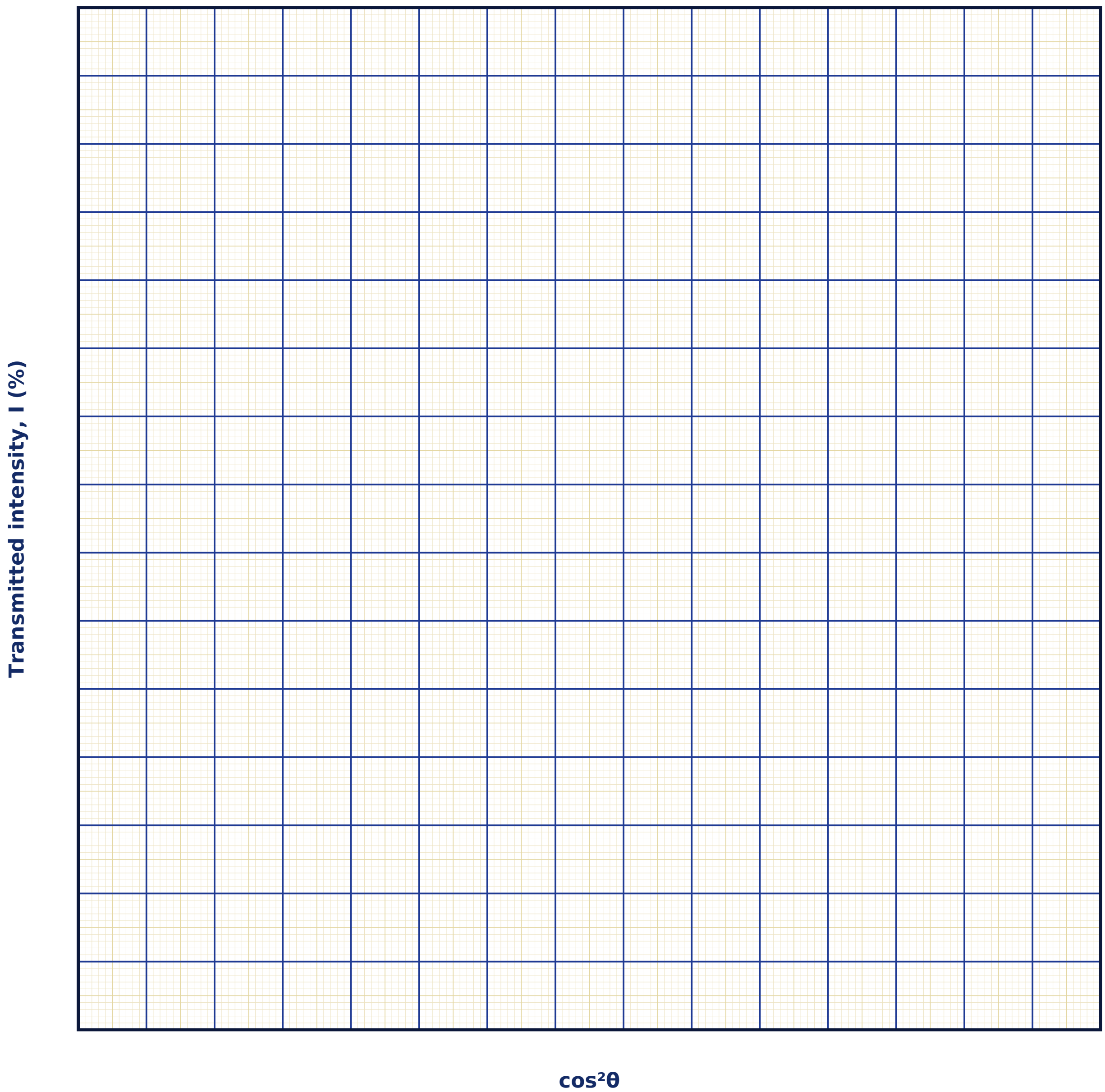
Observation Table

S. No.	θ (°)	$\cos^2\theta$	I observed (%)	$I_0 \cos^2\theta$ calculated (%)

S. No.	θ (°)	$\cos^2\theta$	I observed (%)	I0 $\cos^2\theta$ calculated (%)

Graph

Transmitted Intensity vs $\cos^2\theta$



Calculation

Result

I0 set (%)	I0 from graph slope (%)	% deviation

Maximum Permissible Error

Formula: $\Delta I / I = \Delta I_0/I_0 + 2 \tan\theta \cdot \Delta\theta$

$I = I_0\cos^2\theta$, so $d(\ln I) = d(\ln I_0) - 2 \tan\theta \cdot d\theta$; the angle term grows rapidly as $\theta \to 90°$, so avoid taking your reading very close to the crossed-polarizer position.

Quantity	Measured Value	Least Count (LC)
I■ — Reference intensity (%)		
θ — Analyzer angle (°)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does natural (unpolarized) light not show a $\cos^2\theta$ dependence when passed through a single polarizer alone?

2. What is the difference between a polarizer and an analyzer in this experimental arrangement?

3. What happens to the transmitted intensity when the analyzer is rotated to 90° with respect to the polarizer?

4. Why does unpolarized light lose exactly half its intensity after passing through the first (ideal) polarizer?

5. Distinguish between polarization by selective absorption (as in a Polaroid) and polarization by reflection.

6. At what angle between polarizer and analyzer axes is the rate of change of transmitted intensity greatest?

7. How would you experimentally verify that a given beam of light is completely unpolarized?

4. Dispersion by a Prism

Aim

To study how a prism separates white light into its constituent colours due to the wavelength dependence of refractive index (normal dispersion).

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\mu = \sin[(A+\delta m)/2] / \sin(A/2)$$

μ = refractive index of the prism material
A = apex (prism) angle
δm = angle of minimum deviation

Labelled diagram of the experimental apparatus:

Observation Table

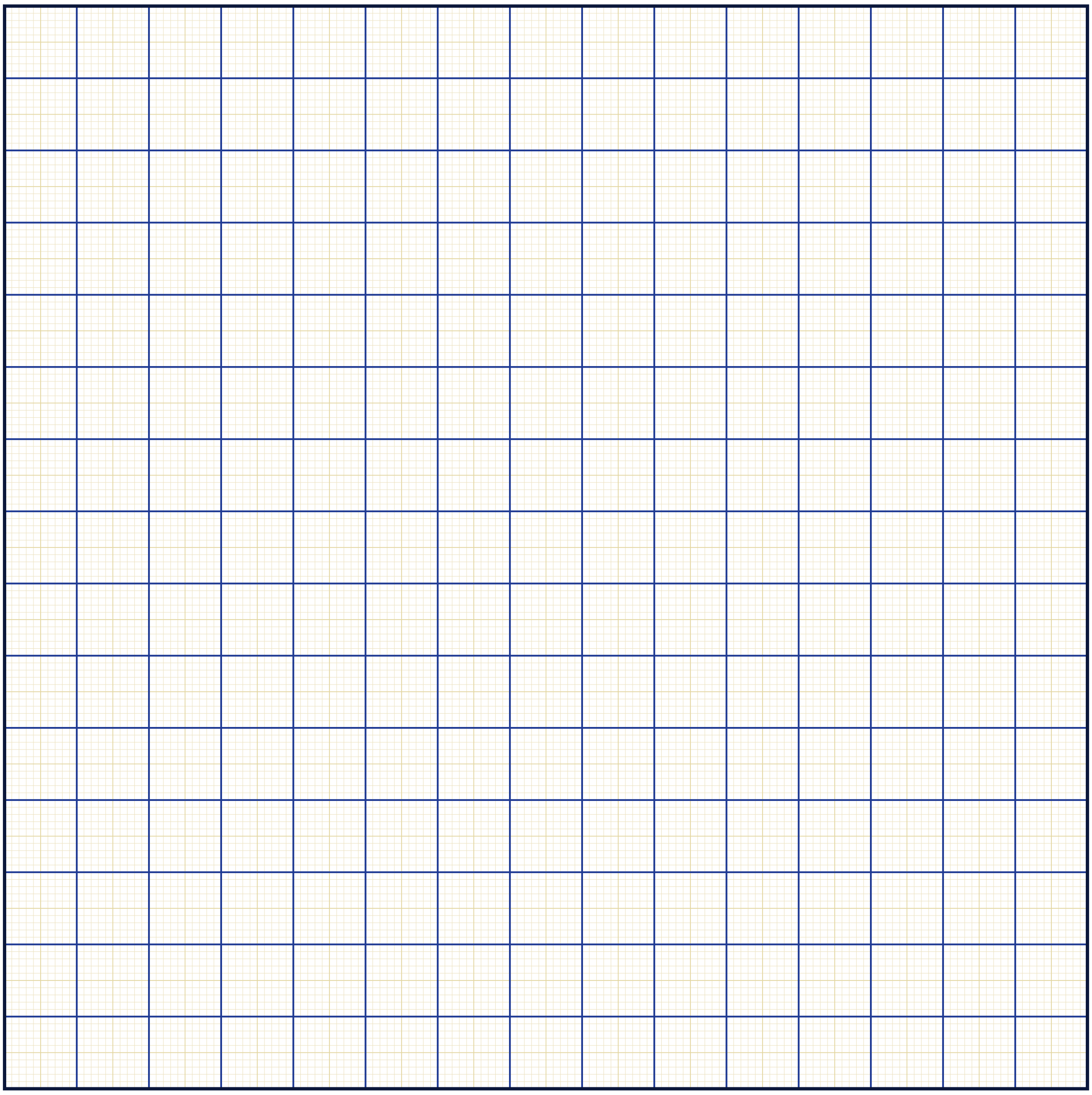
S. No.	Apex angle A (°)	Dispersive power setting	Qualitative spectral spread (narrow/medium/wide)

S. No.	Apex angle A (°)	Dispersive power setting	Qualitative spectral spread (narrow/medium/wide)

Graph

Refractive Index vs $1/\lambda^2$ (Cauchy plot)

Refractive index, n



$1 / \lambda^2$ (μm^{-2})

Calculation

Result

λ (nm)	n(λ) calculated	Colour

Maximum Permissible Error

Formula: $\Delta n \approx \frac{1}{2}\cdot\cot[(A+\delta m)/2]\cdot(\Delta A+\Delta\delta m) + \frac{1}{2}\cdot\cot(A/2)\cdot\Delta A$

From $n = \sin[(A+\delta m)/2] / \sin(A/2)$, where A is the prism apex angle and δm the angle of minimum deviation. Both are read off the same spectrometer circular scale, so use its vernier least count for both ΔA and Δδm.

Quantity	Measured Value	Least Count (LC)
A — Apex angle (°)		
δm — Angle of minimum deviation (°)		
LC — Spectrometer vernier least count (°)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does a prism, unlike a plane mirror, produce a spectrum from white light?

2. What is meant by 'normal' dispersion, and under what conditions does 'anomalous' dispersion occur?

3. Why does the angle-of-minimum-deviation method give a more reliable value of refractive index than an arbitrary angle of incidence?

4. Why is the refractive index of a prism material higher for violet light than for red light?

5. What is meant by the dispersive power of a prism, and how is it calculated?

6. How would the angle of deviation change if the prism material were replaced by one of higher refractive index?

7. Why must the incident beam be effectively monochromatic when measuring refractive index by the minimum-deviation method?

5. Michelson Interferometer

Aim

To study the circular fringe pattern produced by a Michelson interferometer and to use fringe counting to measure the displacement of a movable mirror.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$\Delta L = \Delta m \cdot \lambda / 2$

ΔL = displacement of the movable mirror
Δm = number of fringes shifted, counted at the detector
λ = wavelength of the laser light

Labelled diagram of the experimental apparatus:

Observation Table

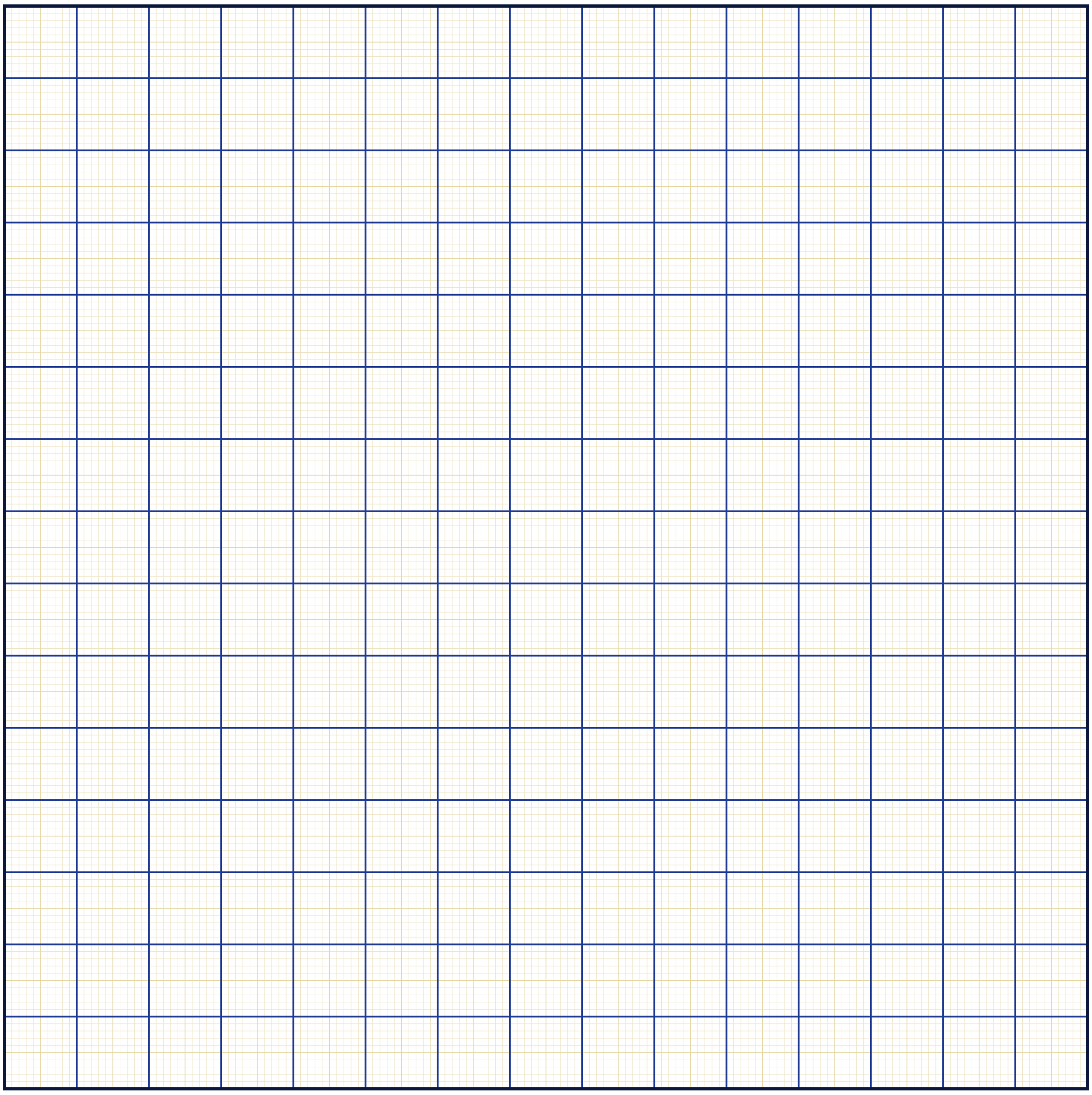
S. No.	ΔL (nm)	λ (nm)	OPD = $2\Delta L$ (nm)	$\Delta m = 2\Delta L/\lambda$ (observed)

S. No.	ΔL (nm)	λ (nm)	OPD = 2ΔL (nm)	Δm = 2ΔL/λ (observed)

Graph

Fringe-Order Shift vs Mirror Displacement

Fringe order shift, Δm



Mirror displacement, ΔL (nm)

Calculation

Result

λ set (nm)	λ from slope (nm)	% error

Maximum Permissible Error

Formula: $\Delta(\Delta L) = (\lambda/2)\times$(fringe-counting uncertainty) $+ \Delta L\times(\Delta\lambda/\lambda)$

$\Delta L = \Delta m\cdot\lambda/2$. The dominant error is usually miscounting fringes by ±1 near the start/end of a sweep, contributing $\pm\lambda/2$ directly; the source's own spectral uncertainty adds a smaller relative term.

Quantity	Measured Value	Least Count (LC)
λ — Wavelength (nm)		
Δλ — Source spectral uncertainty (nm)		
Δm — Fringes counted (fringes)		
± — Counting uncertainty (fringes)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does the fringe pattern in a Michelson interferometer appear as concentric circles rather than straight lines?

2. How is this instrument related historically to the Michelson–Morley experiment on the ether?

3. Why does displacing the movable mirror by a distance λ/2 shift the fringe pattern by exactly one fringe?

4. What is the role of the beamsplitter in this interferometer, and why must it be set at 45° to the incident beam?

5. How can a Michelson interferometer be used to measure the wavelength of an unknown monochromatic source?

6. What change would you expect in the fringe pattern if a thin glass plate were inserted in one arm of the interferometer?

7. Why is mechanical and thermal stability important while performing this experiment?

6. Focusing of Light by a Convex Lens

Aim

To study ray-optics focusing by a convex lens and to determine the diffraction-limited minimum spot size set by the lens aperture.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$r \approx 1.22\ \lambda f / D$$

r = radius of the diffraction-limited focal spot (Airy disc)
λ = wavelength of light
f = focal length of the convex lens
D = diameter of the collimated beam entering the lens (aperture)

Labelled diagram of the experimental apparatus:

Observation Table

S. No.	f (cm)	D (cm)	f-number = f/D	λ (nm)	Spot radius 1.22λf/D (μm)

S. No.	f (cm)	D (cm)	f-number = f/D	λ (nm)	Spot radius 1.22λf/D (μm)

Graph

Spot Radius vs f-number

Spot radius, r (μm)

f-number, N_f = f/D

Calculation

Result

λ set (nm)	λ from slope (nm)	% deviation

Maximum Permissible Error

Formula: $\Delta r / r = \Delta\lambda/\lambda + \Delta f/f + \Delta D/D$

$r = 1.22\lambda f/D$ is a product/quotient of three measured quantities — sum the fractional (least count / value) errors of each.

Quantity	Measured Value	Least Count (LC)
λ — Wavelength (nm)		
f — Focal length (cm)		
D — Aperture diameter (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why can a lens never focus light to a true geometric point, however well it is corrected for aberrations?

2. How does the diffraction-limited spot size influence the achievable precision in a laser-drilling or laser-cutting process?

3. Why can a lens never focus a perfectly collimated beam to a true mathematical point, however good its quality?

4. How does the diffraction-limited spot size change if the lens aperture D is increased?

5. What is meant by the f-number (f/D) of a lens system, and how does it affect the focal spot size?

6. Why does a shorter wavelength give a smaller diffraction-limited spot for the same lens and aperture?

7. Distinguish between a diffraction-limited and an aberration-limited focal spot.

7. Image Formation by a Compound Lens System

Aim

To trace image formation through two thin lenses in series and to determine the total magnification of the compound system.

Date performed: ____________________   Simulator panel used: ______________________________

Formula Used

$1/v - 1/u = 1/f$ (each lens); $m_{total} = m_1 \times m_2$

v = image distance from the lens
u = object distance from the lens
f = focal length of that lens
m_1, m_2 = magnification produced by Lens 1 and Lens 2 respectively

Labelled diagram of the experimental apparatus:

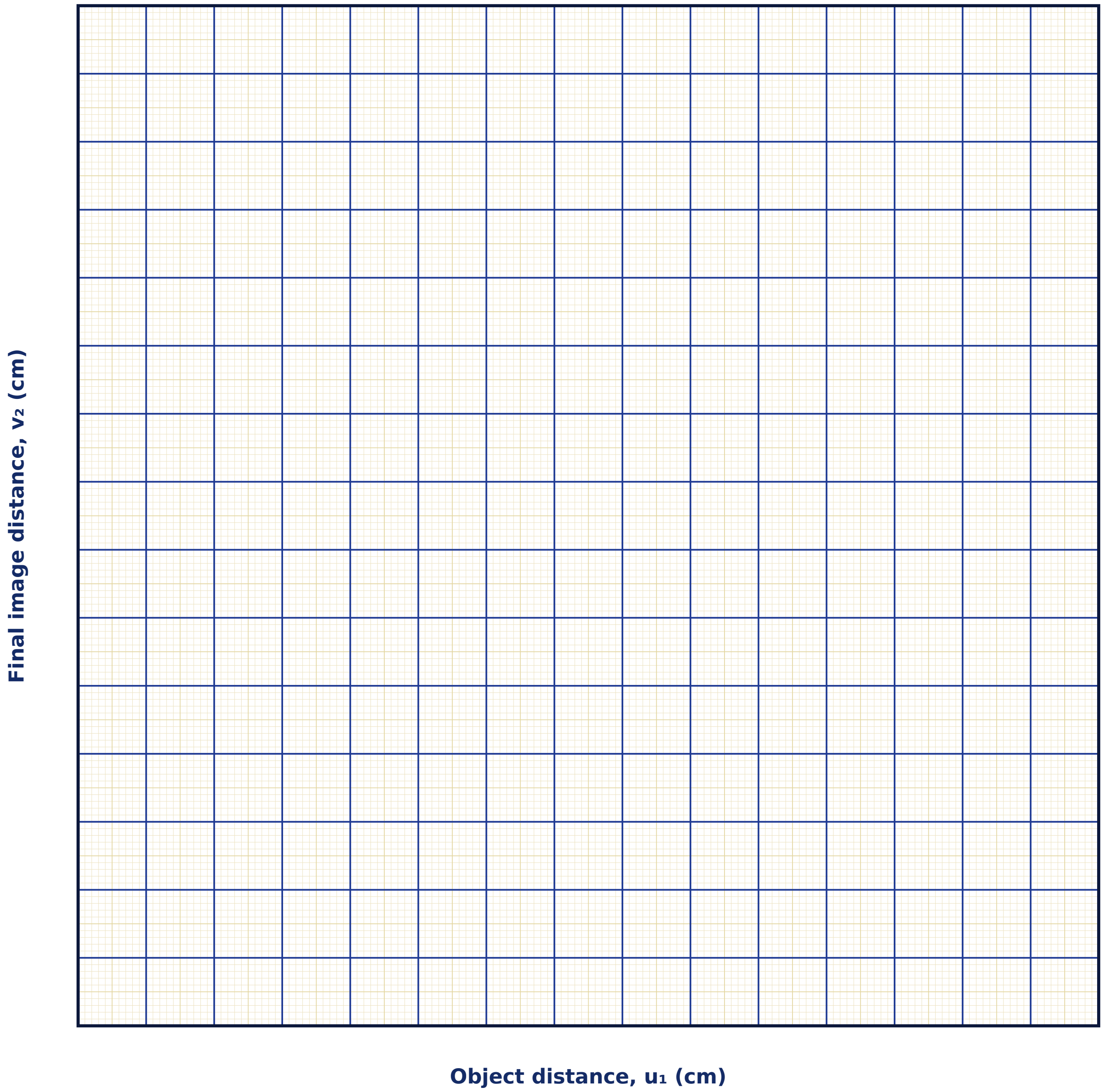
Observation Table

S. No.	u1 (cm)	f1 (cm)	L (cm)	f2 (cm)	v1 (cm)	u2 (cm)	v2 (cm)	m_total

S. No.	u1 (cm)	f1 (cm)	L (cm)	f2 (cm)	v1 (cm)	u2 (cm)	v2 (cm)	m_total

Graph

Final Image Distance vs Object Distance



Calculation

Result

Row	v1 (calculated)	v1 (simulator)	v2 (calculated)	v2 (simulator)	m_total

Maximum Permissible Error

Formula: $\Delta v = v^2 \cdot (\Delta f/f^2 + \Delta u/u^2)$

From $1/v = 1/f + 1/u$ for a single lens stage, propagate the errors in the measured object distance u and the known/measured focal length f. Apply once per lens stage.

Quantity	Measured Value	Least Count (LC)
u — Object distance (cm)		
f — Focal length (cm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. What does it mean physically for lens 2 to receive a 'virtual object'?

2. How does this two-lens analysis extend to a compound microscope with an objective and an eyepiece?

3. Why is the image formed by the first lens treated as the object for the second lens?

4. Under what condition does the total magnification of a two-lens system become negative?

5. How would the final image position change if the separation between the two lenses were increased?

6. What is the difference between linear (transverse) magnification and angular magnification?

7. Name one real optical instrument that relies on a compound lens system similar to this experiment.

8. The Photoelectric Effect

Aim

To study the photoelectric effect and verify Einstein's photoelectric equation by measuring the stopping voltage as a function of incident photon energy for different metals.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$eV_0 = h\nu - \varphi$$

V_0 = stopping potential
h = Planck's constant
ν = frequency of the incident light
φ = work function of the photocathode material
e = charge on the electron

Labelled diagram of the experimental apparatus:

Observation Table

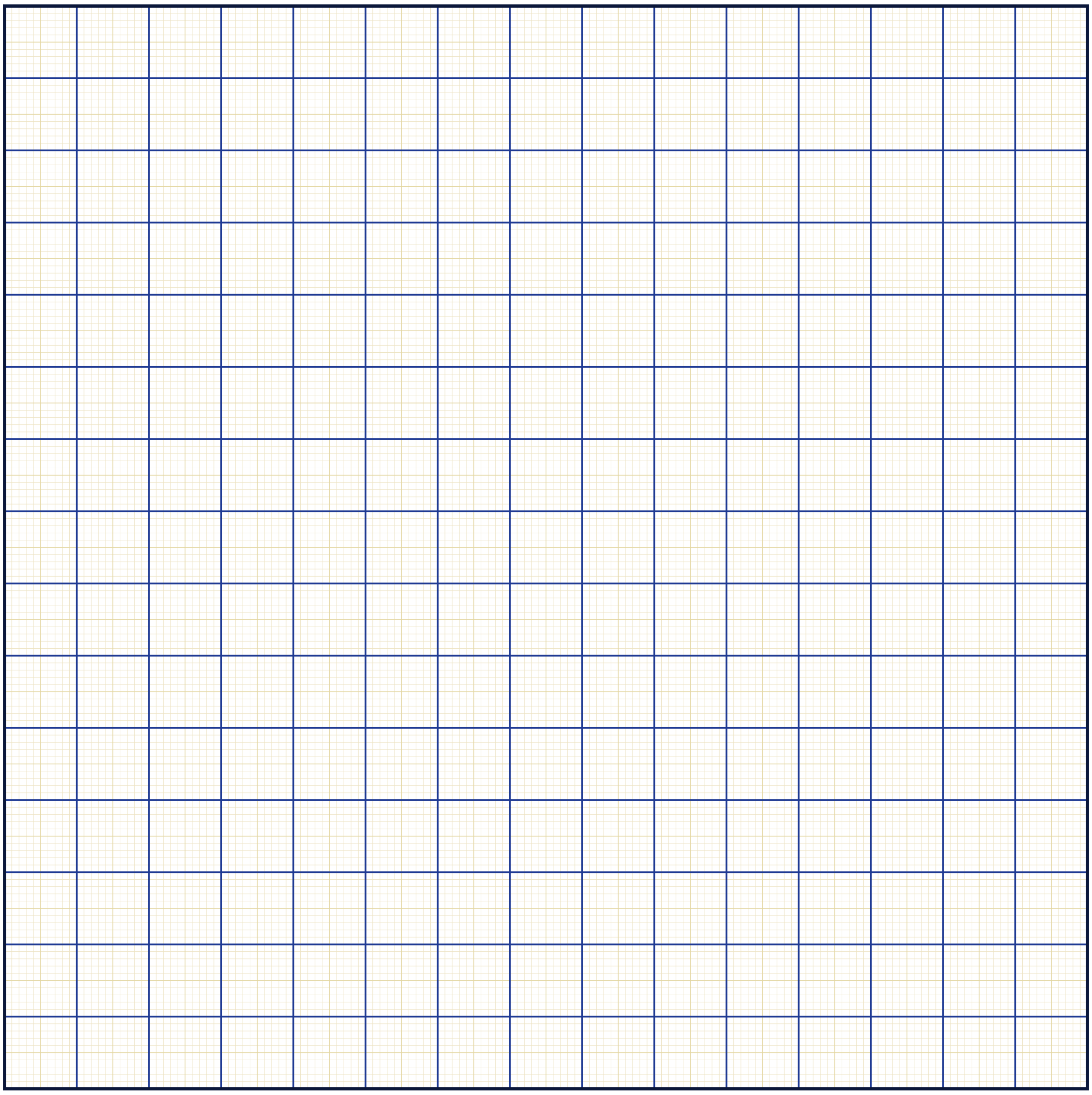
S. No.	Metal	φ (eV)	λ (nm)	$\nu = c/\lambda$ ($\times 10^{14}$ Hz)	$E = h\nu$ (eV)	$V0 = (E-\varphi)/e$ (V)

S. No.	Metal	φ (eV)	λ (nm)	$\nu = c/\lambda$ ($\times 10^{14}$ Hz)	$E = h\nu$ (eV)	$V0 = (E-\varphi)/e$ (V)

Graph

Stopping Voltage vs Frequency

Stopping voltage, V_0 (V)



Frequency, ν ($\times 10^{14}$ Hz)

Calculation

Result

Metal	φ set (eV)	φ from graph (eV)	h/e from slope	% deviation

Maximum Permissible Error

Formula: $\Delta V_0 = (h \cdot \Delta\nu)/e + \Delta\varphi/e$, $\Delta\nu/\nu = \Delta\lambda/\lambda$

$V_0 = (h\nu - \varphi)/e$ is an affine (not multiplicative) relation, so errors from frequency (via the wavelength selector's least count) and from the work function uncertainty simply add.

Quantity	Measured Value	Least Count (LC)
λ — Wavelength (nm)		
ϕ — Work function (eV)		
Δϕ — Work function uncertainty (eV)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why could the photoelectric effect not be explained by the classical wave picture of light?

2. What is the physical significance of the universal slope h/e obtained from the V0–ν graph?

3. Why does photoelectric current not begin until the incident frequency exceeds a threshold value, regardless of intensity?

4. How is the work function of a metal determined from a graph of stopping potential versus frequency?

5. Why does increasing the light intensity increase the photocurrent but not the maximum kinetic energy of the photoelectrons?

6. What was significant about Einstein's explanation of the photoelectric effect that the classical wave theory of light could not account for?

7. What is the physical significance of the slope of the V_0 versus ν graph?

9. Compton Scattering

Aim

To verify the Compton scattering formula by studying the wavelength shift of X-ray photons scattered by a free electron as a function of scattering angle.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$\Delta\lambda = \lambda_C (1 - \cos\theta)$$

Δλ = wavelength shift of the scattered photon
λC = Compton wavelength of the electron (= 2.426 pm)
θ = scattering angle of the photon

Labelled diagram of the experimental apparatus:

Observation Table

S. No.	θ (°)	1 − cosθ	λ (pm)	Δλ (pm) observed	λ' = λ+Δλ (pm)

S. No.	θ (°)	1 − cosθ	λ (pm)	Δλ (pm) observed	λ' = λ+Δλ (pm)

Graph

Wavelength Shift vs (1 − cosθ)

Wavelength shift, Δλ (pm)

1 − cosθ

Calculation

Result

λC (accepted, pm)	λC (from slope, pm)	% error

Maximum Permissible Error

Formula: $\Delta(\Delta\lambda) = \lambda_C \cdot \sin\theta \cdot \Delta\theta$

$\Delta\lambda = \lambda_C(1-\cos\theta)$; differentiating with respect to the scattering angle gives this term. The Compton wavelength λ_C is a fundamental constant with negligible uncertainty.

Quantity	Measured Value	Least Count (LC)
θ — Scattering angle (°)		
Δθ — Goniometer least count (°)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does Compton scattering require the photon to be assigned a definite momentum $p = h/\lambda$?

2. Why is the Compton shift independent of the scattering material?

3. Why is the Compton wavelength shift independent of the wavelength of the incident photon?

4. Why is the Compton effect readily observed with X-rays but not with visible light?

5. What does the Compton wavelength of the electron physically represent?

6. Why does the scattered photon lose energy while the recoiling electron gains kinetic energy in this process?

7. How does Compton scattering provide evidence for the particle (photon) nature of light?

10. Single-Photon Double-Slit Build-up

Aim

To observe the statistical build-up of an interference pattern from individually detected photons sent through a double slit one at a time, demonstrating wave–particle duality.

Date performed: ___________________ Simulator panel used: _____________________________

Formula Used

$$P(y) \propto \cos^2(\pi\, d\, y / \lambda D)$$

P(y) = probability of detecting a photon at position y on the screen
d = separation between the two slits
λ = wavelength of the light
D = slit-to-screen distance

Labelled diagram of the experimental apparatus:

Observation Table

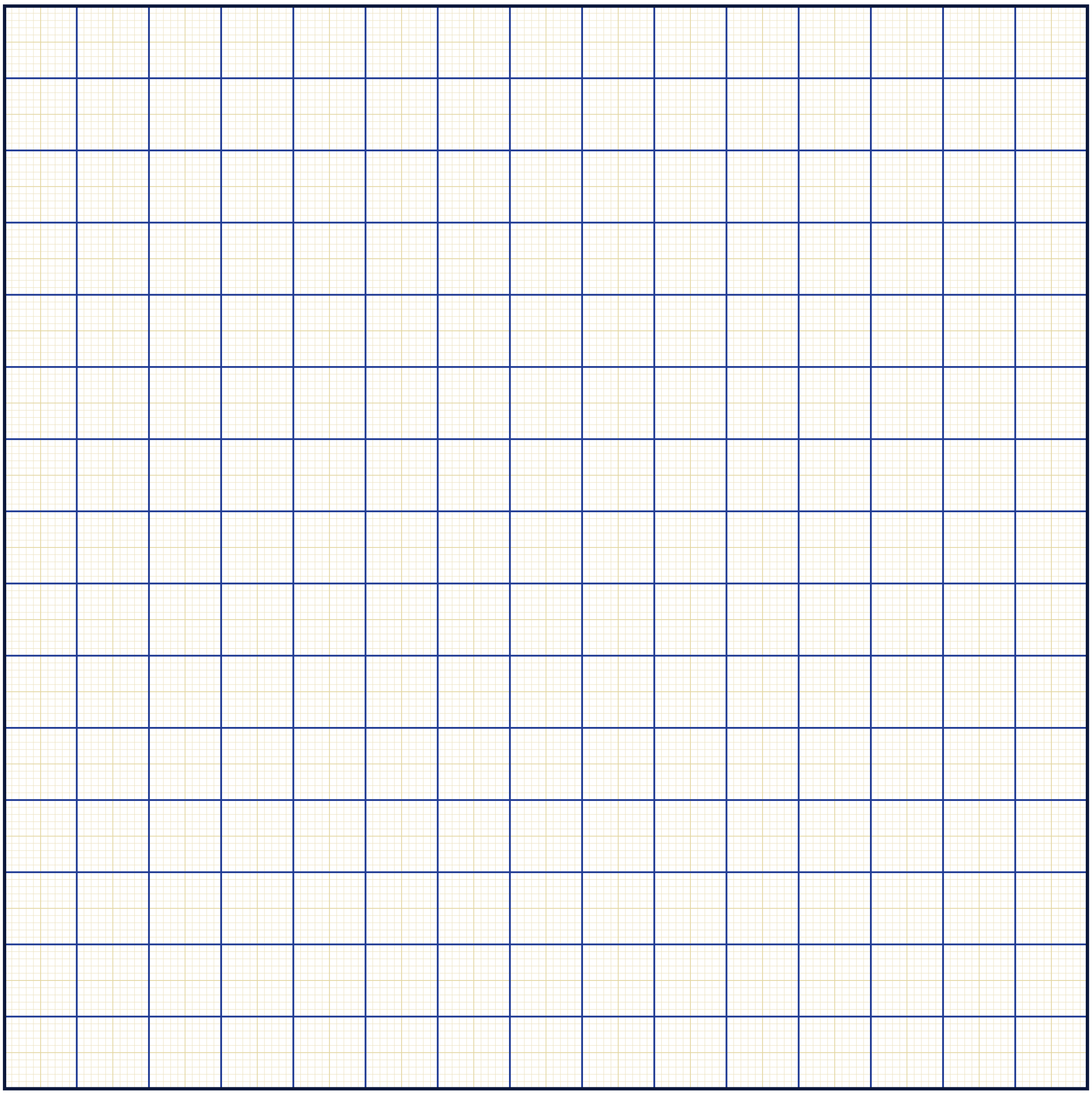
S. No.	Total photons detected	Pattern visibility (none / faint / clear fringes)

S. No.	Total photons detected	Pattern visibility (none / faint / clear fringes)

Graph

Detected Counts vs Position (build-up)

Photon counts, N



Screen position, y (a.u.)

Calculation

Result

Photon count	Estimated visibility V

Maximum Permissible Error

Formula: $\Delta N = \sqrt{N}$ (Poisson counting statistics) → relative error = $1/\sqrt{N}$

This experiment is a qualitative demonstration, not an instrumental measurement — its natural 'error' is the statistical counting uncertainty of accumulating discrete photon detection events, which shrinks as more photons are collected.

Quantity	Measured Value	Least Count (LC)
N — Total photons detected (counts)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Does an individual photon pass through one slit or both? What does this experiment tell us about that question?

2. How does this experiment settle the wave-versus-particle debate about the nature of light?

3. How can an interference pattern build up even when photons are sent through the apparatus one at a time?

4. What happens to the interference pattern if a detector is placed at one slit to determine which slit each photon passed through?

5. What does this experiment demonstrate about the wave–particle duality of light?

6. Why can we not predict exactly where a single photon will land on the screen, only its probability distribution?

7. How does the accumulated pattern from many single photons compare with the classical double-slit fringe pattern?

11. Coherence Length & Coherence Time

Aim

To study how the spectral linewidth of a source limits its coherence time and coherence length, and to observe the resulting loss of fringe visibility with increasing path difference.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$L_c = c \cdot \tau_c \approx c / \Delta\nu$$

L_c = coherence length
c = speed of light
τ_c = coherence time
$\Delta\nu$ = frequency bandwidth (linewidth) of the source

Labelled diagram of the experimental apparatus:

Observation Table

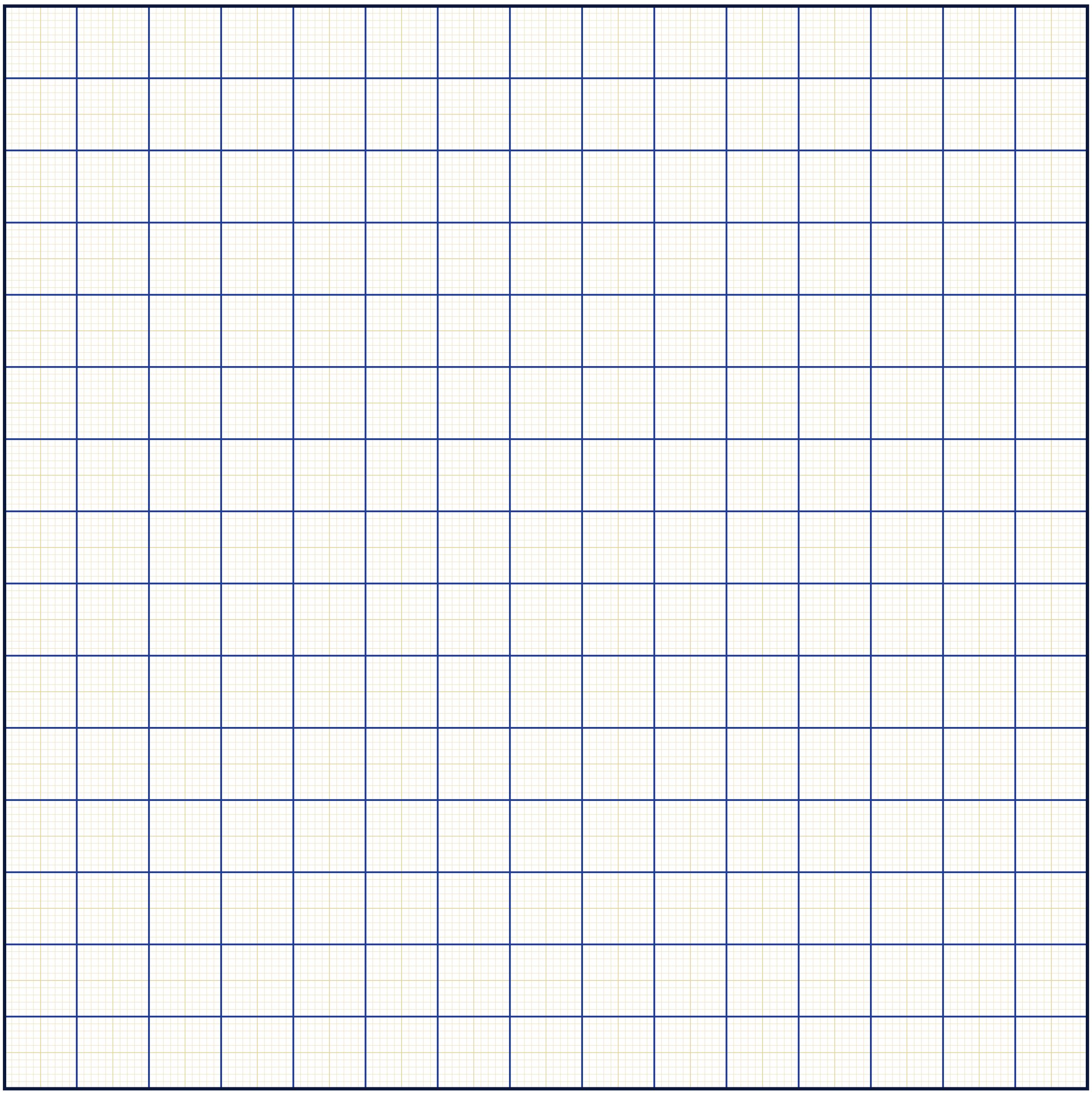
S. No.	Source	$\Delta\nu$	τ_c (calculated)	$L_c = c\tau_c$ (calculated)	Path diff. at half-visibility (observed)

S. No.	Source	$\Delta\nu$	τ_c (calculated)	$L_c = c\tau_c$ (calculated)	Path diff. at half-visibility (observed)

Graph

Coherence Length vs Spectral Linewidth

Coherence length, L_c (m, log scale suggested)



Linewidth, $\Delta\nu$ (Hz, log scale suggested)

Calculation

Result

Source	Lc (calculated)	Lc (from visibility curve)	Remarks

Maximum Permissible Error

Formula: $\Delta L_c / L_c = \Delta(\Delta\nu) / \Delta\nu$

$L_c = c/\Delta\nu$ (c is exact), so the fractional error in coherence length equals the fractional error in your measurement of the source's spectral linewidth.

Quantity	Measured Value	Least Count (LC)
$\Delta\nu$ — Spectral linewidth (MHz)		
$\Delta(\Delta\nu)$ — Linewidth measurement uncertainty (MHz)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why can two separate light bulbs never produce a stationary interference pattern, however carefully arranged?

2. How is coherence length exploited (or limited) in holography and in optical coherence tomography?

3. Why do white-light sources produce fringes only very close to zero path difference, unlike a laser?

4. How is coherence time τc related to the spectral bandwidth $\Delta\nu$ of a source?

5. Distinguish between temporal coherence and spatial coherence.

6. Why does an LED have a much shorter coherence length than a single-mode laser?

7. How would you experimentally estimate the coherence length of a source using an interferometer such as Michelson's?

12. Directionality & Beam Quality

Aim

To compare the angular divergence of a laser beam with that of a conventional light source and to determine the spot size produced at a given propagation distance.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$D_{spot}(L) \approx D_0 + \theta_{div} \cdot L$$

D_{spot} = beam spot diameter after travelling a distance L
D_0 = initial beam diameter at the source
θ_{div} = beam divergence (full angle)
L = propagation distance

Labelled diagram of the experimental apparatus:

Observation Table

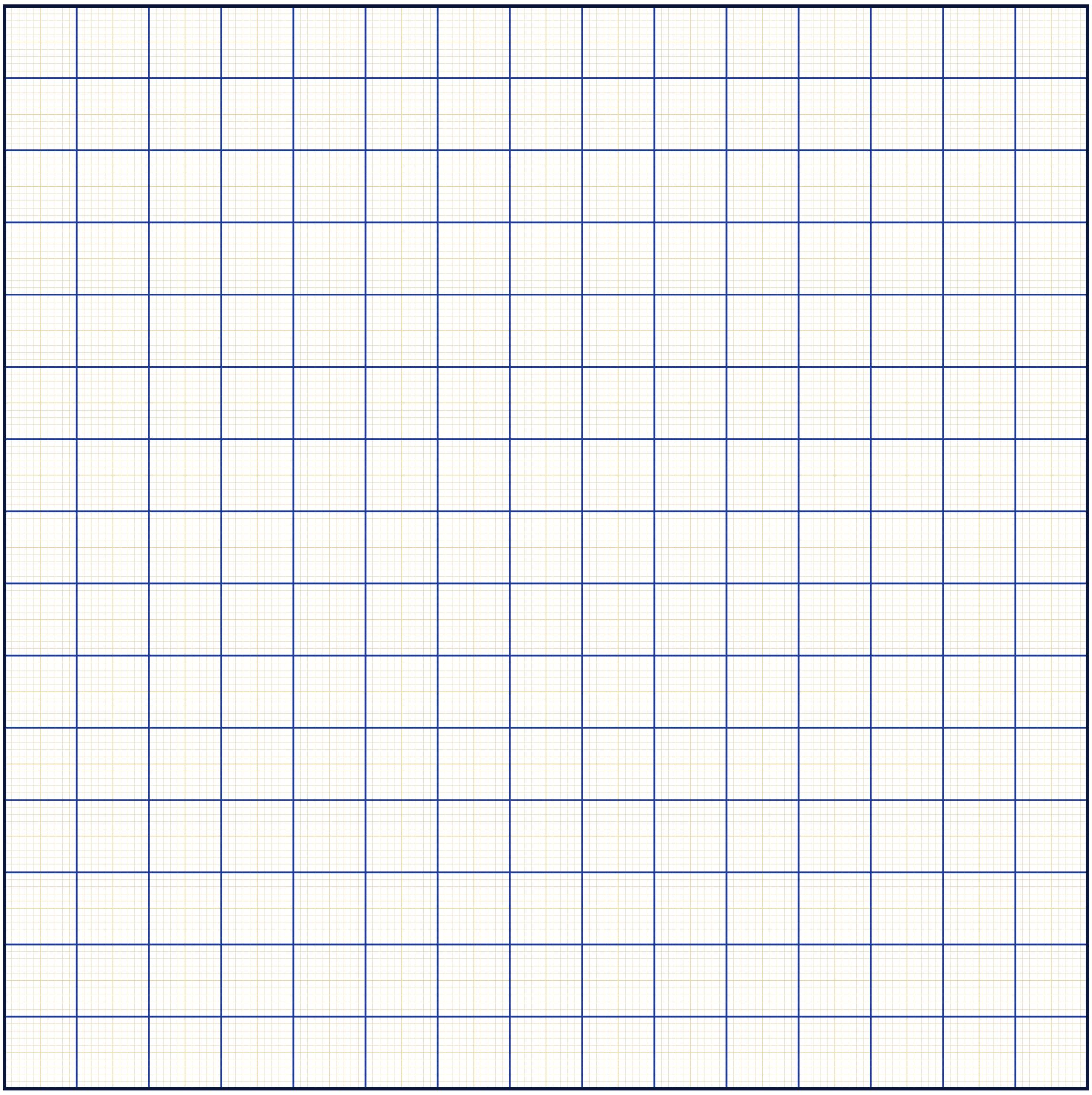
S. No.	θ_div (mrad)	L (m)	Spot diameter (observed)

S. No.	θ_div (mrad)	L (m)	Spot diameter (observed)

Graph

Spot Diameter vs Propagation Distance

Spot diameter, D_spot (mm)



Propagation distance, L (m)

Calculation

Result

θ_div set (mrad)	θ_div from slope (mrad)	% deviation

Maximum Permissible Error

Formula: $\Delta D\_spot = \Delta D_0 + L \cdot \Delta\theta\_div + \theta\_div \cdot \Delta L$

$D\_spot(L) = D_0 + \theta\_div \cdot L$ is a sum, so absolute (not fractional) errors add directly, each weighted by the relevant partial derivative.

Quantity	Measured Value	Least Count (LC)
D■ — Initial beam diameter (mm)		
θ_div — Angular divergence (mrad)		
L — Propagation distance (m)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why is directionality considered one of the defining properties of laser light, alongside coherence and monochromaticity?

2. How does beam divergence limit the maximum useful range of a laser rangefinder or LIDAR system?

3. Why does an ordinary lamp's light spread out so much more than a laser beam over the same distance?

4. What is meant by beam divergence, and in what units is it typically expressed?

5. How does the M^2 (beam quality) factor describe how close a real laser beam is to an ideal Gaussian beam?

6. Why is low divergence important for applications such as laser range-finding or optical communication?

7. How would you measure the divergence angle of a laser beam experimentally?

13. Monochromaticity & Spectral Purity

Aim

To compare the spectral purity of different light sources by measuring the ratio Δλ/λ of linewidth to central wavelength.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

Purity = Δλ / λ

Δλ = spectral linewidth of the source
λ = central wavelength of the source

Labelled diagram of the experimental apparatus:

Observation Table

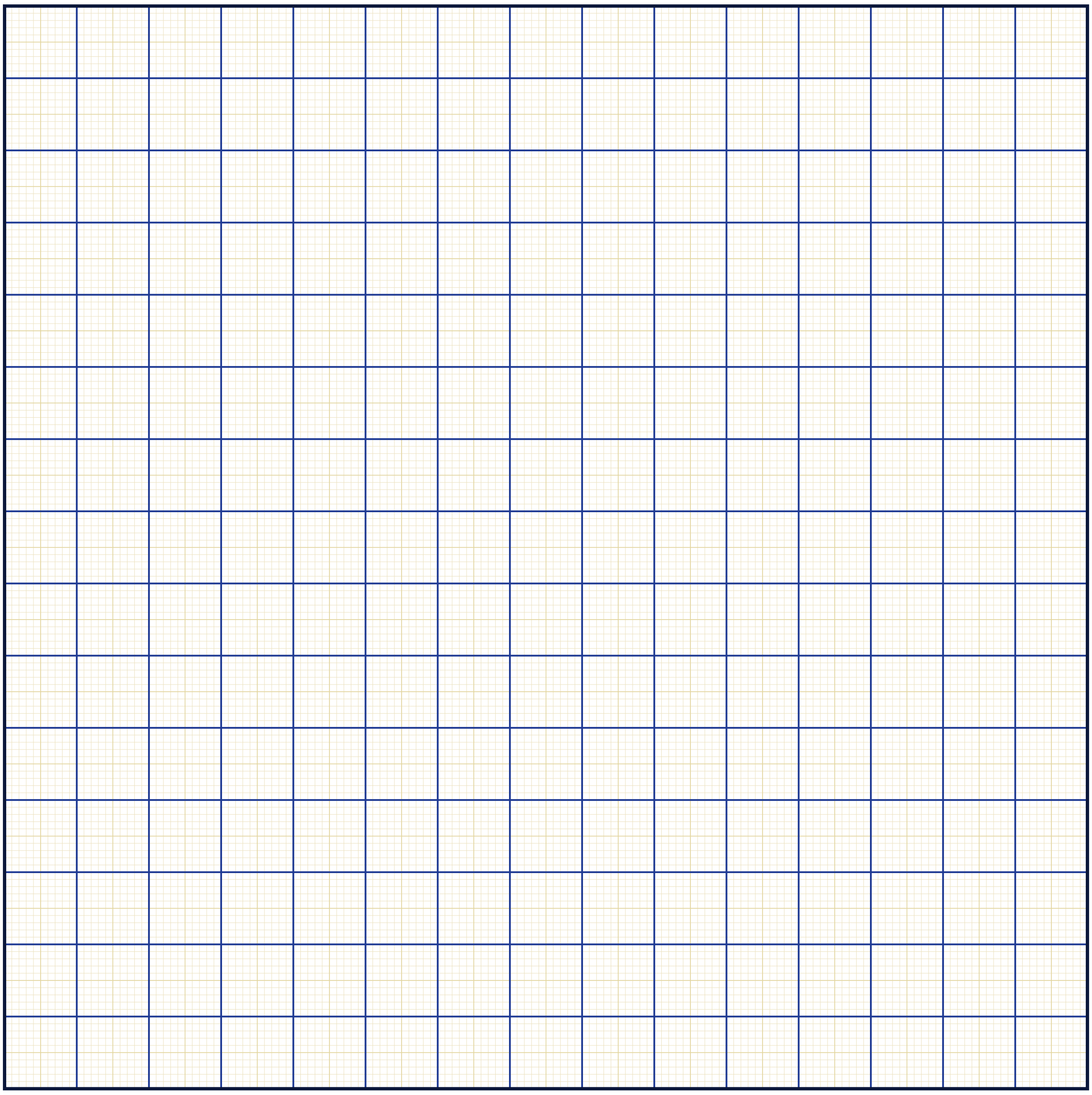
S. No.	Source type	λ (nm)	Δλ (nm)	Purity Δλ/λ

S. No.	Source type	λ (nm)	Δλ (nm)	Purity Δλ/λ

Graph

Spectral Purity Comparison

Purity, Δλ/λ (log scale suggested)



Source type (categorical)

Calculation

Result

Source	$\Delta\lambda/\lambda$	Relative ranking

Maximum Permissible Error

Formula: $\Delta(\text{purity}) / \text{purity} = \Delta W/W + \Delta\lambda/\lambda$

purity = W/λ, where W is the measured spectral linewidth. Sum the fractional error in the linewidth measurement and in the central-wavelength measurement.

Quantity	Measured Value	Least Count (LC)
W — Spectral linewidth (nm)		
λ — Central wavelength (nm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why is high spectral purity essential for wavelength-division multiplexing in optical fibre communication?

2. How are spectral purity and coherence length related to one another physically?

3. What does a smaller value of $\Delta\lambda/\lambda$ indicate about the spectral purity of a source?

4. Why is a single-mode laser far more monochromatic than a filtered lamp of the same nominal wavelength?

5. How is the spectral linewidth $\Delta\lambda$ of a source related to its coherence length?

6. Why is high spectral purity important for applications such as spectroscopy or interferometry?

7. What instrument would you use to directly measure the linewidth of a laser source?

14. Making a Hologram

Aim

To understand the recording and reconstruction stages of a hologram and to appreciate the role of coherent laser light in capturing both amplitude and phase information.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$I = |E_{obj} + E_{ref}|^2$$

I = intensity recorded at the photographic (holographic) plate
E_obj = object-beam field (carries amplitude & phase of the object)
E_ref = reference-beam field

Labelled diagram of the experimental apparatus:

Observation Table

(Conceptual experiment — record your step-by-step observations/sketches below.)

Graph

(Qualitative experiment — use this space to sketch the process/stages.)

Calculation

Result

Stage	Physical process

Maximum Permissible Error

Holography as presented here is a qualitative recording/reconstruction demonstration, not an instrumental measurement — there is no single numeric result with a propagated maximum permissible error. Use the space below to sketch/note the four stages instead.

Viva-Voce Questions

1. Why must the reference and object beams originate from the same laser source?

2. What distinguishes a hologram from an ordinary photograph in terms of the information each one records?

3. Why must both the object beam and the reference beam originate from the same laser source?

4. What information is recorded on a holographic plate that is not recorded on an ordinary photograph?

5. Why must a holographic plate record both amplitude and phase information for a full 3-D reconstruction?

6. What happens if the reconstruction beam does not match the original reference beam in wavelength or direction?

7. Explain the difference between the real and virtual images formed during hologram reconstruction.

15. Red → Ultraviolet Wavelength & Photon-Energy Explorer

Aim

To explore, across the visible spectrum and into the ultraviolet, how wavelength, frequency, photon energy and ocular/skin hazard all change together as a single continuous slider is swept from deep red toward UV-C.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$E = h\nu = hc / \lambda$

E = photon energy
h = Planck's constant
c = speed of light
λ = wavelength of the light

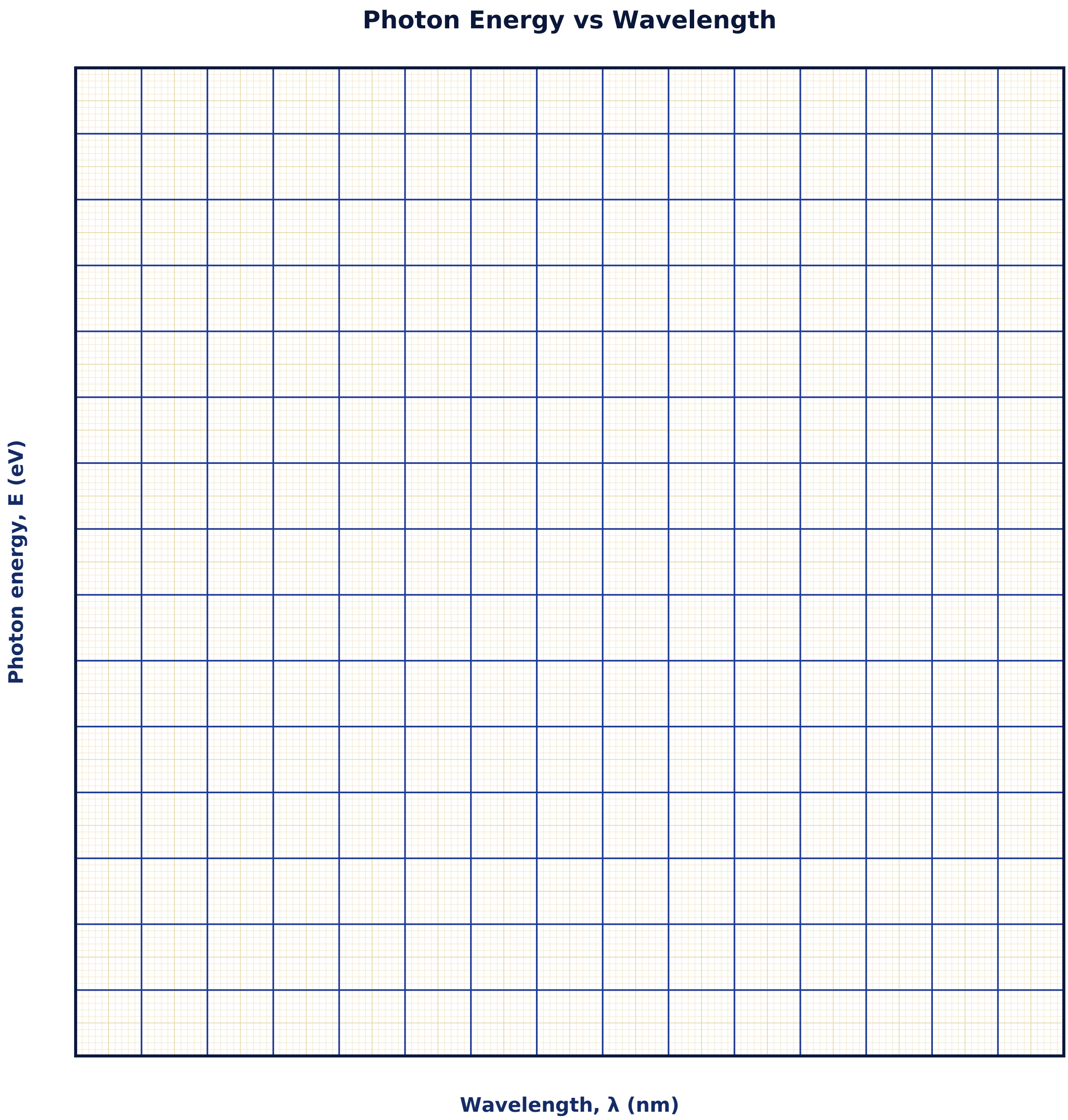
Labelled diagram of the experimental apparatus:

Observation Table

S. No.	Wavelength λ (nm)	Region	Frequency ν (THz)	Photon energy E (eV)	Hazard note (summary)

S. No.	Wavelength λ (nm)	Region	Frequency ν (THz)	Photon energy E (eV)	Hazard note (summary)

Graph



Calculation

Result

λ (nm)	E calculated (eV)	E displayed by simulator (eV)	Region

Maximum Permissible Error

Formula: $\Delta E / E = \Delta\lambda / \lambda$

$E = hc/\lambda$ (h, c exact), so the fractional error in photon energy equals the fractional error in the wavelength measurement/setting.

Quantity	Measured Value	Least Count (LC)
λ — Wavelength (nm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does photon energy increase as wavelength decreases, even though the speed of light c is constant?

2. Why is an invisible ultraviolet or infrared beam potentially more hazardous than a visible beam of the same power?

3. At what approximate wavelength does human vision stop, and what governs this limit?

4. Give one real laser type each that would fall in the UV-A, UV-B and UV-C regions covered by this explorer.

5. Why is the relationship between photon energy and frequency linear, while the relationship between energy and wavelength is not?

6. Why are X-rays and gamma rays, despite their very short wavelengths, considered more hazardous to biological tissue than visible light?

7. What determines whether a given photon can ionize an atom it interacts with?

16. Laser Doppler Vibrometry

Aim

To study how the Doppler shift imposed on a laser beam reflected from a vibrating surface encodes the surface's instantaneous velocity, and to see why this makes possible a completely non-contact vibration sensor.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$f_D = 2v \cos\theta / \lambda \; ; \; v_{max} = A\omega$$

v = instantaneous velocity of the vibrating surface
θ = angle of incidence of the probe beam
A, ω = vibration amplitude and angular frequency

Labelled diagram of the experimental apparatus:

Observation Table

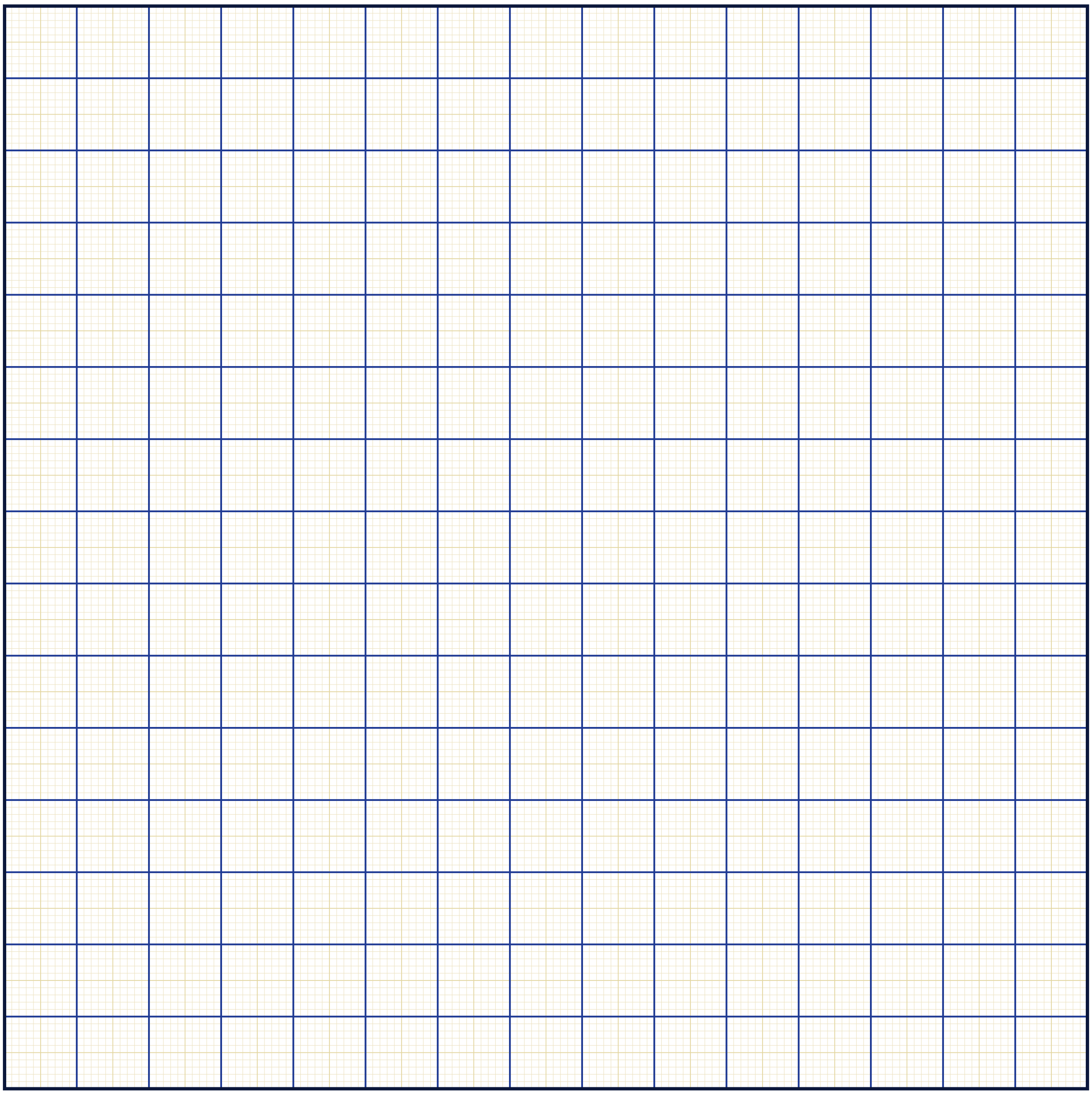
S. No.	A (µm)	θ (°)	f (Hz)	$v_{max} = A\omega$ (mm/s)	$f_{D,max}$ (kHz)

S. No.	A (µm)	θ (°)	f (Hz)	v_max = Aω (mm/s)	f_D,max (kHz)

Graph

Peak Doppler Shift vs Vibration Frequency

Peak Doppler shift, f_D,max (kHz)



Vibration frequency, f (Hz)

Calculation

Result

Quantity	Mean value	Unit

Maximum Permissible Error

Formula: $\Delta v_{max} / v_{max} = \Delta A/A + \Delta f/f$

$v_{max} = A\omega = A(2\pi f)$ is a simple product of the vibration amplitude and (angular) frequency, so their fractional errors simply add.

Quantity	Measured Value	Least Count (LC)
A — Vibration amplitude (µm)		
f — Vibration frequency (Hz)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does the Doppler shift depend on cosθ rather than simply on the surface's speed?

2. Explain why heterodyne (beat-frequency) detection, rather than simply measuring reflected intensity, is necessary to recover the Doppler shift.

3. How would the measured signal change if the surface vibrated with two superposed frequencies instead of one pure tone?

4. Why is velocity, rather than displacement, the more natural quantity for this technique to measure directly?

5. Give one practical application where a contact sensor would be unsuitable but an LDV would work well, and explain why.

6. What would happen to the recovered signal if the target surface were optically rough versus mirror-smooth?

17. Optical Fiber — Numerical Aperture & Attenuation

Aim

To determine the Numerical Aperture and acceptance angle of a step-index optical fiber from its core and cladding refractive indices, and to study how signal power decays exponentially with distance due to fiber attenuation.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$NA = \sqrt{n_1^2 - n_2^2} \; ; \; P_{out} = P_{in} \cdot 10^{(-\alpha L/10)}$$

n_1, n_2 = core and cladding refractive indices
α = attenuation coefficient of the fiber
L = fiber length

Labelled diagram of the experimental apparatus:

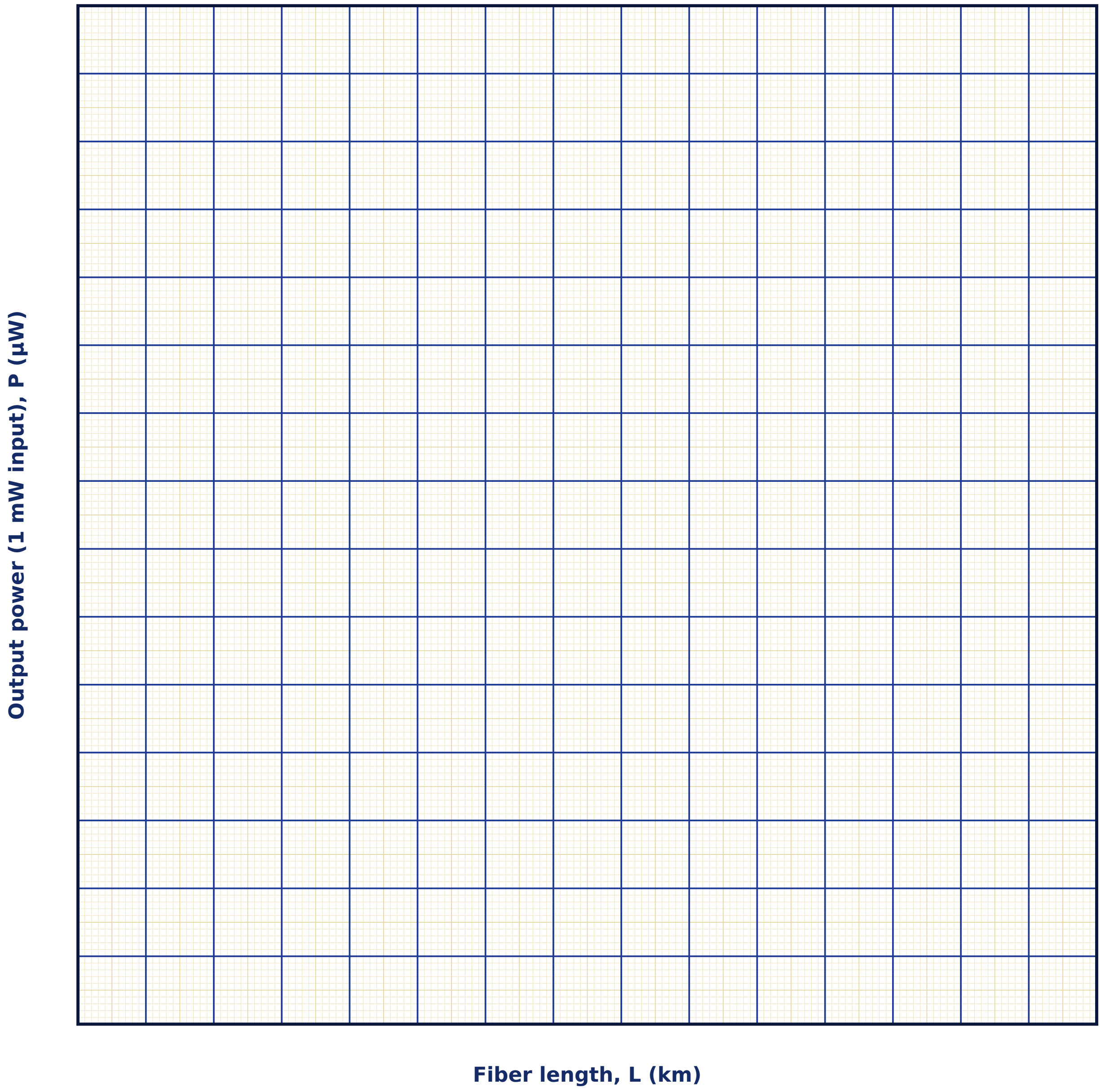
Observation Table

S. No.	n_1	n_2	NA	α (dB/km)	L (km)	P_out (μW, 1 mW input)

S. No.	n_1	n_2	NA	α (dB/km)	L (km)	P_out (μW, 1 mW input)

Graph

Output Power vs Fiber Length



Calculation

Result

Quantity	Mean value	Unit

Maximum Permissible Error

Formula: $\Delta(NA) \approx (n_1\Delta n_1 + n_2\Delta n_2) / NA$

$NA = \sqrt{(n_1^2 - n_2^2)}$; since NA is a square root of a difference of squares, its absolute error is dominated by whichever refractive index (core or cladding) carries the larger measurement uncertainty, weighted by that index's own value.

Quantity	Measured Value	Least Count (LC)
n■ — Core refractive index ((dimensionless))		
n■ — Cladding refractive index ((dimensionless))		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why must the core refractive index always exceed the cladding index for a fiber to guide light at all?

2. Explain the trade-off between a large Numerical Aperture (easy light-launching) and modal dispersion in a real multimode fiber.

3. Why is fiber attenuation conventionally expressed in decibels per kilometre rather than as a simple fraction lost per kilometre?

4. From the formula $P_{out} = P_{in} \cdot 10^{(-\alpha L/10)}$, derive the fiber length at which exactly half the input power remains, in terms of α.

5. Give one practical reason why single-mode fibers (very small core, small NA) are preferred over multimode fibers for very long-distance telecommunications links.

6. How would you expect the attenuation coefficient α to depend on the wavelength of light used, physically?

18. Laser Speckle Metrology — Non-Contact Surface Roughness

Aim

To study how the grain size and contrast of a laser speckle pattern depend on the illumination geometry and on the roughness of the scattering surface, and to see how this forms the basis of a non-contact surface-finish inspection technique.

Date performed: ____________________ Simulator panel used: ______________________________

Formula Used

$$s = 1.22\lambda z/D \;;\; \text{Contrast} \approx 1 - \exp[-(2\pi\sigma/\lambda)^2]$$

z = distance from the illuminated spot to the observation screen
D = diameter of the illuminated beam/aperture
σ = RMS surface roughness

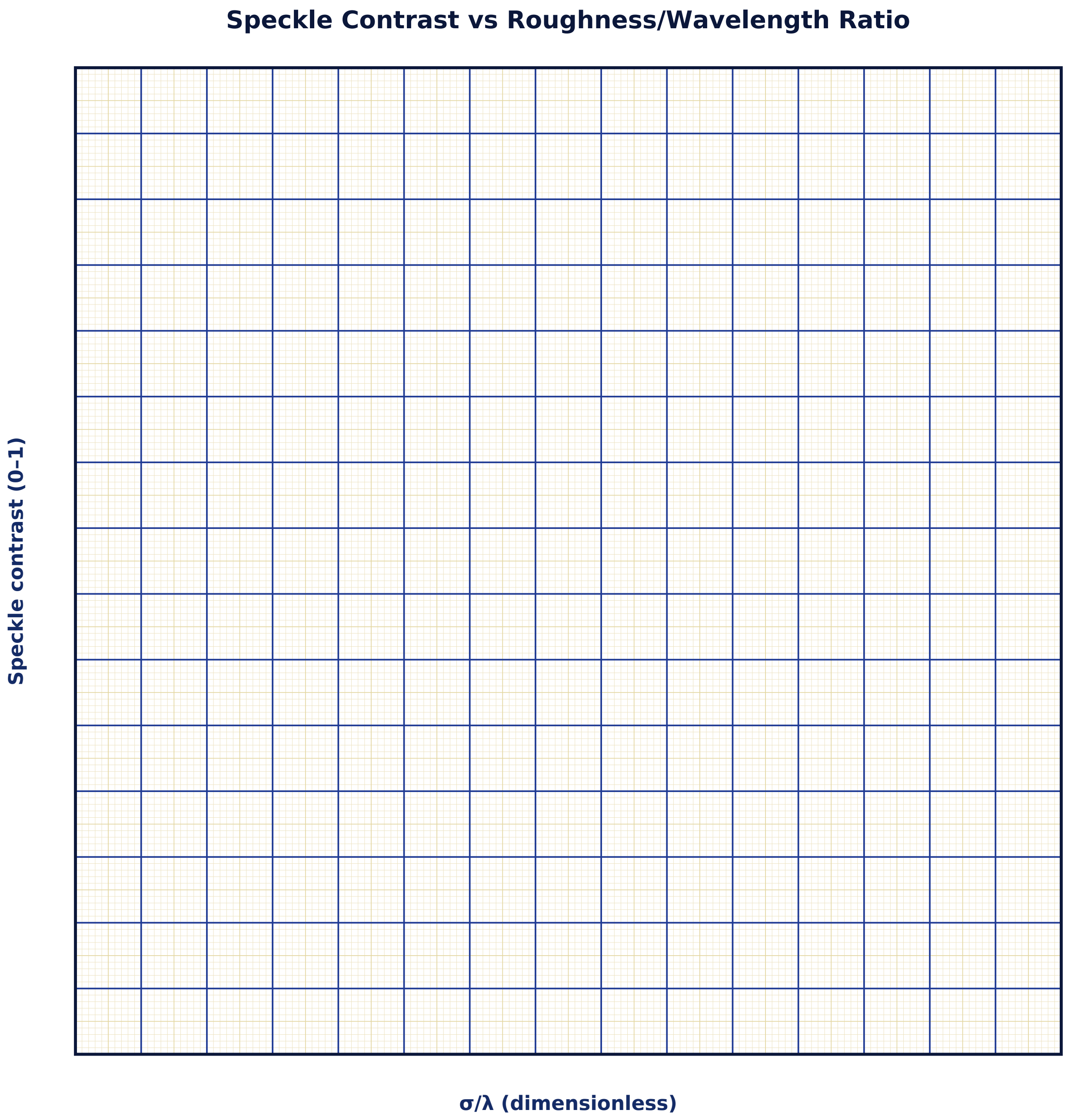
Labelled diagram of the experimental apparatus:

Observation Table

S. No.	D (mm)	z (cm)	σ (nm)	σ/λ	Contrast, C

S. No.	D (mm)	z (cm)	σ (nm)	σ/λ	Contrast, C

Graph



Calculation

Result

Quantity	Mean value	Unit

Maximum Permissible Error

Formula: $\Delta s / s = \Delta z/z + \Delta D/D$

The speckle grain size $s = 1.22\lambda z/D$ is a product/quotient of measured quantities (λ taken as exact for a stabilized laser source), so the fractional errors in the screen distance z and beam diameter D simply add.

Quantity	Measured Value	Least Count (LC)
z — Screen distance (cm)		
D — Illuminated beam diameter (mm)		

Calculation of maximum permissible error:

Viva-Voce Questions

1. Why does speckle contrast depend on surface roughness while speckle grain size does not?

2. Explain physically why illuminating a rougher surface produces a higher-contrast speckle pattern.

3. How would using a shorter-wavelength laser (e.g., a blue laser instead of a red He-Ne) change the sensitivity of this technique to a given surface's roughness?

4. Why is a genuinely coherent source essential for observing speckle, and what would happen with an ordinary incoherent lamp instead?

5. Give one advantage of a laser-speckle-based surface inspection method over a traditional contact stylus profilometer, beyond simply avoiding scratches.

6. What practical factors (other than roughness) might also affect the measured speckle contrast in a real industrial setting?